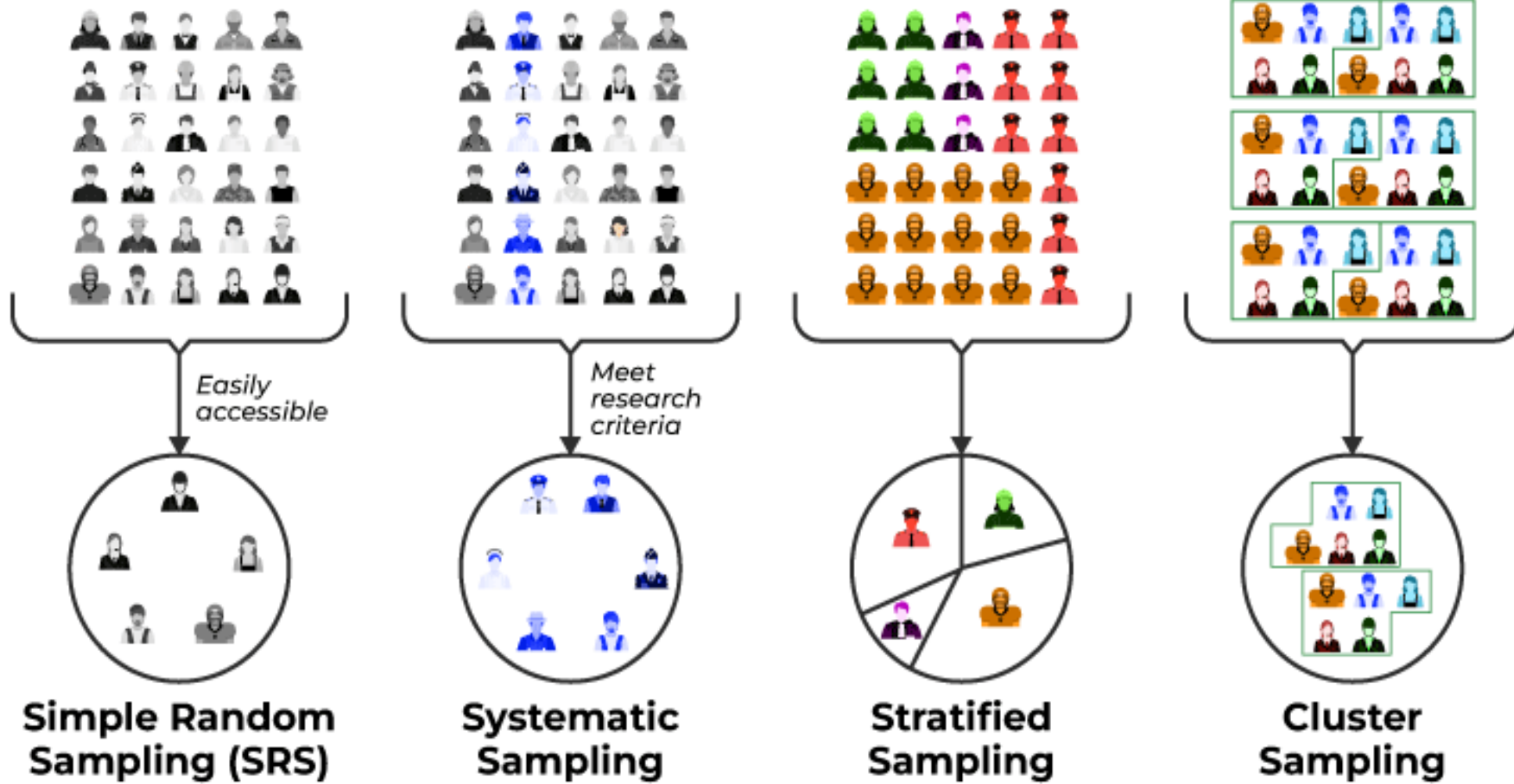


PROBABILITY SAMPLING METHODOLOGIES



1. Simple random sampling

- ❖ This method of sample selection is the easiest to understand. Each participant has an equal chance of taking part in the study using this strategy. Each member of this sample population has an equal chance of being chosen at random to become an object.
- ❖ It is one of the most basic and commonly used forms of probability sampling.
- ❖ For Example, if a university dean wanted to gather input from students about how they felt about the professors and their level of education, this sample could include all 1000 students at the university. To create this sample, 100 students can be chosen at random from any class.

Steps in Simple Random Sampling

- ❖ **Define the Population:**
 - ❖ Clearly identify the population you want to study.
 - ❖ Example: All 1,000 students in a school.
- ❖ **Assign Numbers to Each Member:**
 - ❖ Assign a unique identifier (e.g., student ID or a sequential number) to every individual in the population.

❖ **Select the Sample Randomly:**

- ❖ Use a random selection method, such as:
- ❖ Random number generators
- ❖ Lottery methods (drawing names or numbers from a hat)

❖ **Collect Data from the Selected Sample:**

- ❖ Only the randomly selected individuals are surveyed or analyzed.

Advantages of Simple Random Sampling

- ❖ **Unbiased:** Every individual has an equal chance of selection, reducing selection bias.
- ❖ **Easy to Implement:** Simple and straightforward with modern tools (e.g., random number generators).

Disadvantages of Simple Random Sampling

- ❖ **Not Always Practical:** It may be challenging to list every individual in a large or dispersed population.
- ❖ **May Not Guarantee Representation:** Random selection could lead to an unbalanced sample (e.g., more males than females).

2. Cluster sampling

- ❖ A sample technique called cluster sampling divides the respondent population into equal clusters.
- ❖ Cluster sampling is a sampling method where the population is divided into groups, called clusters, and then entire clusters are randomly selected to collect data from all members of the chosen clusters. This is especially useful when the population is large and geographically disseminated.
- ❖ **Steps in Cluster Sampling**
- ❖ **Divide the Population into Clusters:**
 - ❖ Divide the entire population into non-overlapping, naturally occurring groups or clusters (e.g., schools, neighborhoods, or districts).
 - ❖ Clusters should ideally represent the diversity of the population
- ❖ **Select Clusters Randomly:**
 - ❖ Use a random sampling method to select a few clusters. You do not need to select every cluster.
- ❖ **Collect Data:**
 - ❖ Data is collected from all individuals within the selected clusters

3. Systematic sampling

- ❖ By using systematic sampling, a population is sampled by randomly selecting respondents at equal intervals. Picking a beginning point and then choosing responders at a predetermined sample interval is the method for choosing the sample.
- ❖ Systematic Sampling is a probability sampling method where you select every k -th individual from a population list, starting from a randomly chosen starting point. This method is simpler and quicker than simple random sampling, especially for large populations.
- ❖ As an diagram, when choosing 1,000 volunteers for the Olympics from a list of 10,000 applicants, each applicant is assigned a count between 1 and 10,000. Then a sample of 1,000 volunteers can be obtained by counting backwards from 1 and selecting each respondent with a 10-second interval.

Advantages of Systematic Sampling

- ❖ Simplicity: Easier and faster to implement than simple random sampling.
- ❖ Efficiency: Works well with ordered populations, such as lists or production lines.
- ❖ Ensures Even Coverage: By spreading the sample evenly across the population, systematic sampling can prevent clustering.

4. Stratified random sampling

- ❖ In the research design phase, stratified random sampling is a technique for segmenting the respondent population into discrete but pre-defined parameters. The responders in this method don't overlap; rather, they speak for the entire population as a whole.
- ❖ Stratified random sampling is a probability sampling method where the population is divided into distinct subgroups, called strata, based on shared characteristics.
- ❖ A random sample is then taken from each stratum to ensure that the sample is representative of the population as a whole
- ❖ For Example, a researcher examining persons from various socioeconomic backgrounds can identify respondents based on their yearly earnings. Afterward, some of the objects from these samples can be employed for the research study. This creates smaller groups of persons or samples.